Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates 2^t or 2^{-t} to obtain $\pm A\ln 2 \times 2^{\pm t}$	AO1.1a	M1	
	Obtains $\frac{dy}{dt} = (\pm A\ln 2)2^t$ and $\frac{dx}{dt} = (\pm B\ln 2)2^{-t}$	AO1.1b	A1	$\frac{dy}{dt} = (3\ln 2)2^t$ $\frac{dx}{dt} = (-4\ln 2)2^{-t}$ $\frac{dy}{dx} = \frac{(3\ln 2)2^t}{(-4\ln 2)2^{-t}}$ $= -\frac{3}{4} \times 2^{2t}$
	Uses chain rule with correct $\frac{dy}{dt}$ and $\frac{dx}{dt}$ and completes rigorous argument to obtain fully correct printed answer	AO2.1	R1	
(b)	Rearranges to write 2^{-t} in terms of x or 2^t in terms of y Or Writes given expression in terms of t	AO3.1a	M1	$2^t = \frac{y+5}{3}$ $2^{-t} = \frac{x-3}{4}$ $1 = \left(\frac{y+5}{3}\right)\left(\frac{x-3}{4}\right)$ $12 = xy + 5x - 3y - 15$ $xy + 5x - 3y = 27$
	Eliminates t Or Compares coefficients PI by $a = 5$ or $b = -3$	AO1.1a	M1	
	Completes rigorous argument to obtain correct values of a, b and c and write the Cartesian equation in the required form ISW	AO2.1	R1	**ALT** $xy + ax + by = (4 \times 2^{-t} + 3)(3 \times 2^t - 5) + a(4 \times 2^{-t} + 3) + b(3 \times 2^t - 5)$ $= 12 - 15 + (4a - 20)2^{-t} + (3b + 9)2^t + 3a - 5b$ $a = 5$, $b = -3$ $xy + 5x - 3y = -3 + 15 + 15$ $= 27$
				Total 6 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Selects appropriate technique to differentiate	AO3.1a	M1	$2(x+y-2)\left(1+\frac{dy}{dx}\right)=e^y\frac{dy}{dx}$
Differentiates term involving e^y correctly	AO1.1b	B1	$\frac{dy}{dx}=0 \Rightarrow x+y-2=0$
Differentiates fully correctly	AO1.1b	A1	$\Rightarrow 0 = e^y - 1$
Uses $\frac{dy}{dx}=0$	AO1.1a	M1	$y = 0$ $x = 2$
Eliminates x or y from the equation of the curve	AO1.1a	M1	
Obtains correct y CAO	AO1.1b	A1	
Obtains correct x CAO	AO1.1b	A1	
			Total 7 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Finds the difference between the maximum and minimum values of y	AO3.1b	M1	$x^2 + 2xy + 2y^2 = 10$ $2x+2y+2x\frac{dy}{dx}+4y\frac{dy}{dx}=0$
Uses implicit differentiation	AO1.1a	M1	
Differentiates correctly	AO1.1b	A1	Highest and lowest points occur when $\frac{dy}{dx}=0$
States stationary points occur when $\frac{dy}{dx}=0$	AO2.4	R1	$\frac{dy}{dx}=0 \Rightarrow x=-y$
Uses $\frac{dy}{dx}=0$ to find x in terms of y (or vice versa)	AO1.1a	M1	$y^2 - 2y^2 + 2y^2 = 10$ $y=\pm\sqrt{10}$ $\therefore$ Height = $\sqrt{10}-(-\sqrt{10})$
Finds $x = -y$	AO1.1b	A1	= $2\sqrt{10}$ = 6.32m
Deduces maximum and minimum values of y FT 'their' expression provided all M1 marks have been awarded	AO2.2a	A1F	
States the height of the sculpture above the	AO2.2a	A1F	

platform FT 'their' max and min values for y provided all M1 marks have been awarded			
			Total 8 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses a correct method for finding $\frac{dy}{dx}$ evidence for this includes sight of $\frac{dy}{dx}$ or $\frac{dx}{dt}$ and chain rule OR an attempt at implicit or explicit differentiation of a correct Cartesian equation or 'their' equation from part **(b)**	AO1.1a	M1	$\frac{dx}{dt} = 3t^2$ $\frac{dy}{dt} = 2t$ $\frac{dy}{dx} = \frac{2t}{3t^2}$ When $t = -2$ $\frac{dy}{dx} = -\frac{1}{3}$ **ALT** $y = (x-2)^{\frac{2}{3}} - 1$ $\frac{dy}{dx} = \frac{2(x-2)^{-\frac{1}{3}}}{3}$ When $t = -2$, $x = -6$ $\frac{dy}{dx} = \frac{2(-6-2)^{-\frac{1}{3}}}{3} = -\frac{1}{3}$
	Obtains correct $\frac{dy}{dx}$	AO1.1b	A1	
	Substitutes $t = -2$ (or $x = -6$) into 'their' equation for $\frac{dy}{dx}$	AO1.1a	M1	
	Obtains correct simplified gradient of the curve FT 'their' equation for $\frac{dy}{dx}$	AO1.1b	A1F	
(b)	Eliminates t or makes t the subject in one expression (evidence for this includes one equation with t as the subject or two equations with equal powers of t.)	AO1.1a	M1	$t^3 = (x - 2)$, $t^2 = (y + 1)$ $t^6 = (x - 2)^2$, $t^6 = (y + 1)^3$ $(x - 2)^2, = (y + 1)^3$

Finds a correct Cartesian equation in any form	AO1.1b	A1	**ALT** $t^3 = (x - 2)$ $t = (x-2)^{\frac{1}{3}}$ $y = (x-2)^{\frac{2}{3}} - 1$
			Total 6 marks

Q5.

(a) (i) $2x - 2y\dfrac{dy}{dx} = 0$

Correct differentiation

M1

$\dfrac{dy}{dx} = \dfrac{x}{y}$

at (p,q) $\dfrac{dy}{dx} = \dfrac{p}{q}$

(p,q)substituted into correct derivative or $x = p$ $y= q$ stated

AG

A1

2

Alternative:

$y = \sqrt{x^2 - 8}$ $\dfrac{dy}{dx} = \dfrac{1}{2} \times 2x\left(x^2 - 8\right)^{-\frac{1}{2}} = \dfrac{x}{y}$

(M1)

$= \dfrac{p}{q}$

(A1)

(2)

Alternative:

$\dfrac{dy}{dt} = 1 + \dfrac{2}{t^2}$ $\dfrac{dx}{dt} = 1 - \dfrac{2}{t^2}$

Attempt parametric derivatives and use chain rule.

(M1)

$$\frac{dy}{dx} = \frac{1+\frac{2}{t^2}}{1-\frac{2}{t^2}} = \frac{t+\frac{2}{t}}{t-\frac{2}{t}} = \frac{x}{y}$$

at (p,q) $\frac{dy}{dx} = \frac{p}{q}$

(p,q)substituted into correct derivative.

(A1)

(2)

(ii) tangent at (p,q) $y - q = \frac{p}{q}(x - p)$

ACF

B1

tangent at $(p,-q)$ $y - (-q) = \frac{-p}{q}(x - p)$

ACF

B1

add $2y = 0$

Solve tangent equations for y.

M1

conclusion $y = 0 \Rightarrow$ intersect on Ox

Conclusion required

A1

4

Alternative:

tangent at (p,q) $y - q = \frac{p(x-p)}{q}$

ACF

(B1)

tangent at $(p,-q)$ $y - (-q) = \frac{-p(x-p)}{q}$

ACF

(B1)

When $y = 0$ $\frac{-q^2}{p} = x - p$ and $\frac{q^2}{-p} = x - p$

Substitute y = 0 into both candidate's tangents and solve for x.

(M1)

$x = p - \frac{q^2}{p}$ is on both lines, so intersect on x axis

Conclusion

(A1)

$x + y = 2t \quad x - y = \frac{4}{t}$

(4)

(b) $x^2 = t^2 + 4 + \frac{4}{t^2} \quad y^2 = t^2 - 4 + \frac{4}{t^2}$

Attempt to square x and y and subtract.

M1

$x^2 - y^2 = 8$

All correct AG Allow 8 = 8

A1

2

Alternative:

$(x - y)(x + y) = 2t \times \frac{4}{t}$

Attempt to eliminate t

(M1)

$x^2 - y^2 = 8$

(A1)

(2)

[8]

Q6.

(a) $\left(\frac{dx}{dt} =\right) -16e^{-2t} \quad \left(\frac{dy}{dt} =\right) 4e^{2t}$

Both derivatives correct

B1

$\frac{dy}{dx} = \frac{\text{candidate's } \frac{dy}{dt}}{\text{candidate's } \frac{dx}{dt}}$

chain rule used correctly

M1

$\frac{dy}{dx} = \frac{4e^{2t}}{-16e^{-2t}} \quad \left(= -\frac{1}{4}e^{4t}\right)$

Simplification not required $4e^{2t}$ and $-16e^{-2t}$ must be seen. ISW.

A1

3

(b) (i) $t = \ln 2$ gradient at $P = -4$

B0 if ISW result is used.

B1ft

1

(ii) coordinates of P

$x = -2$

B1

$y = 12$

B1

2

(iii) gradient of normal = $\frac{1}{4}$

ft gradient at P

B1ft

equation of normal $\frac{y-12}{x--2} = \frac{1}{4}$

Set up equation of normal

M1

at $y = 0$ $x = -50$

(− 50,0) CSO

A1

3

(c) $xy + 4y - 4x = (8e^{-2t} - 4)(2e^{2t} + 4) + 4(2e^{2t} + 4) - 4(8e^{-2t} - 4)$

Write $xy + 4y - 4x$ in terms of t.

M1

$= 16 + 32e^{-2t} - 8e^{2t} - 16 + 8e^{2t} + 16 - 32e^{-2t} + 16$

Multiply out and simplify using $e^{-2t}e^{2t} = 1$ PI

m1

$(xy + 4y - 4x) = 32$

Correct working to $k = 32$

$k = 32$ NMS; SC1

A1

3

Alternative:

$e^{-2t} = \dfrac{x+4}{8}$ or $e^{2t} = \dfrac{y-4}{2}$

Write e^{-2t} in terms of x or e^{2t} in terms of y.

Condone sign errors

(M1)

$e^{-2t}e^{2t} = \left(\dfrac{x+4}{8}\right)\left(\dfrac{y-4}{2}\right) = \dfrac{xy+4y-4x-16}{16} = 1$

Multiply out and use e^{-2t} $e^{2t} = 1$

(m1)

$xy + 4y - x = 32$

All correct with $k = 32$

(A1)

(3)

[12]

Q7.

(a) $xy^2 + 3y = (8t^2 - t)\left(\dfrac{3}{t}\right)^2 + 3\left(\dfrac{3}{t}\right)$

Substitute and expand

M1

$= 72 - \dfrac{9}{t} + \dfrac{9}{t} = 72$

$k = 72$

A1

2

Alternative

$$x = 8\left(\frac{3}{y}\right)^2 - \frac{3}{y}$$

Eliminate t

(M1)

$xy^2 + 3y = 72$

k = 72

(A1)

(2)

(b) (i) $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 16t - 1 \quad \dfrac{\mathrm{d}y}{\mathrm{d}t} = -\dfrac{3}{t^2}$

B1B1

$$t = \frac{1}{4} \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-\dfrac{3}{\left(\frac{1}{4}\right)^2}}{16 \times \dfrac{1}{4} - 1}$$

Use chain rule $\left(\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{-3}{16t^3 - t^2}\right)$ *and calculate gradient using t =* $\dfrac{1}{4}$

M1

$= -16$

A1

$$t = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}}$$

Calculate x and y using t = $\dfrac{1}{4}$

M1

$x = \dfrac{1}{4} \qquad y = 12$

Both correct

A1

tangent $\quad y = -16x + 16$

ACF CSO

$y - 12 = -16\left(x - \frac{1}{4}\right)$ *ISW*

A1

7

Alternative

$2xy\frac{dy}{dx} + y^2$

Product rule attempted; two terms added, one with $\frac{dy}{dx}$

(M1A1)

$+3\frac{dy}{dx} = 0$

(B1)

$t = \frac{1}{4}$ $x = \frac{8}{16} - \frac{1}{4}$ $y = \frac{3}{\frac{1}{4}}$

Calculate x and y using t = $\frac{1}{4}$

(M1)

$x = \frac{1}{4}$ y = 12

Both correct.

(A1)

$\left(\frac{dy}{dx} = \frac{-y^2}{2xy + 3}\right)$ $\frac{dy}{dx} = -16$

Calculate gradient from candidate's expression.

m1

tangent $y = -16x + 16$

ACF CSO

$y - 12 = -16\left(x - \frac{1}{4}\right)$ *ISW*

(A1)

(7)

Alternative

$x = \frac{72-3y}{y^2}$

Correct expression for x from candidate's implicit equation.

Quotient rule attempted; y^4 and two terms subtracted.

(M1)

$\frac{dx}{dy} = \frac{y^2(-3)-(72-3y)\times 2y}{y^4}$

(A1)

$\left(\frac{dx}{dy} = \frac{3y-144}{y^3}\right)$

Numerator; first term; second term

(A1)

$t = \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}} = 12$

(B1)

$\frac{dx}{dy} = -\frac{1}{16} \quad \frac{dy}{dx} = -16$

Use $t = \frac{1}{4}$ to calculate y

m1

$x = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} = \frac{1}{4}$

Evaluate and invert.

(B1)

$y = -16x + 16$

(A1)

(7)

Alternative for $\frac{dx}{dy}$

Use $t = \frac{1}{4}$ to calculate x

$x = \frac{72}{y^2} - \frac{3}{y}$

ACF CSO

(M1)

$$\frac{dx}{dy} = -\frac{144}{y^3} + \frac{3}{y^2}$$

(A1)

$$\left(\frac{dx}{dy} = \frac{3y - 144}{y^3}\right)$$

Correct expression for x from candidate's implicit equation and attempt derivatives

(A1)

(ii) $y = -16 \times \frac{3}{2} + 16 = -8$

Substitute $x = \frac{3}{2}$ into candidate's tangent; calculate y

M1

$\frac{3}{2}(-8)^2 + 3 \times (-8) = 96 - 24 = 72$

$y = -8$ used to verify 72

A1

2

[11]

Q8.

(a) (i) $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{6\cos 2\theta}{-2\sin\theta}$

condone coefficient errors

M1
A1

$= \frac{6(1 - 2\sin^2\theta)}{-2\sin\theta}$

Use $\cos 2\theta = 1 - 2\sin^2\theta$

m1

$= 6\sin\theta - 3\text{cosec}\theta$

$a = 6$ $b = -3$

A1

4

(ii) $\theta = \frac{\pi}{6}$ $\frac{dy}{dx} = 6 \times \frac{1}{2} - 3 \times 2 = -3$

$\theta = \frac{\pi}{6}$ substituted into their $\frac{dy}{dx}$ and evaluated

B1ft

gradient normal = $\frac{1}{3}$

ft $\frac{dy}{dx}$, provided non-zero

B1ft

2

(b)

$$\left.\begin{aligned} y &= 6\sin\theta\cos\theta \\ &= (\pm)6\sqrt{1-\cos^2\theta} \times \cos\theta \\ &= (\pm)6\sqrt{1-\left(\frac{x}{2}\right)^2} \times \left(\frac{x}{2}\right) \end{aligned}\right\}$$

Correct expansion of sin 2θ and use $x = 2\cos\theta$ to eliminate θ

M1

Correct elimination of θ

A1

$y^2 = \frac{9}{4}x^2(4 - x^2)$

$p = \frac{9}{4}$ OE and $(4 - x^2)$ shown

A1

Alternative: using verification

$y^2 = 9\sin^2 2\theta = 36\sin^2\theta\cos^2\theta$

must be squared

(M1)

$x^2(4 - x^2) = 4\cos^2\theta \times 4\sin^2\theta$

(A1)

$p = \frac{9}{4}$ OE

or $y^2 = \frac{9}{4}x^2(4 - x^2)$

(A1)

3

[9]

Q9.

$9x^2 - 6xy + 4y^2 = 3$

$18x = 0$

= 0 PI

B1

$-6y - 6x\dfrac{dy}{dx}$

or $\dfrac{d(6xy)}{dx} = 6y + 6x\dfrac{dy}{dx}$ *seen separately*

B1

$+ 8y\dfrac{dy}{dx}$

$\dfrac{dy}{dx}(-6x + 8y) = 6y - 18x$

B1

Use $\dfrac{dy}{dx} = 0$

M1

$\Rightarrow y = 3x$ or $x = \dfrac{y}{3}$

CSO

A1

$y = 3x \Rightarrow 9x^2 - 6x \times 3x + 4(3x)^2 = 3$

$27x^2 = 3 \Rightarrow x = \pm\dfrac{1}{3}$ OE

Substitute $y = ax$ into equation and solve for a value of x or y. Condone missing brackets.

m1

Both values of x or y required.
ft on their $y = 3x$

A1ft

$\left(\frac{1}{3},1\right)$ $\left(-\frac{1}{3},-1\right)$

CSO Correct corresponding simplified values of x and y.
Withhold if additional answers given

A1

[8]

Q10.

(a) (i) $\frac{dx}{dt}=3e^{t}$ $\frac{dy}{dt}=2e^{2t}+2e^{-2t}$

Both derivatives attempted and one correct

M1

Both correct

A1

$t=0$ gradient $=\frac{4}{3}$

cso Condone $\frac{dy}{dx}=\frac{4}{3}$

A1

3

(ii) $y\frac{4}{3}(x-3)$ oe

ft on non–zero gradient

B1ft

1

(b) $e^{2t}=\frac{x^2}{9}$ or $9e^{2t}=x^2$ or $e^{t}=\frac{x}{3}$ or $e^{2t}=\left(\frac{x}{3}\right)^2$

or $t=\ln\left(\frac{x}{3}\right)$ or $2t=\ln\left(\frac{x^2}{9}\right)$

M1

$y=\frac{x^2}{9}-\frac{9}{x^2}$

Equation required

A1

2

[6]

Q11.

(a) (i) $\left(\frac{dx}{d\theta}=\right)-6\sin 2\theta,\ \left(\frac{dy}{d\theta}=\right)-2\sin\theta$

$\left(\frac{dx}{d\theta}=\right)p\sin\theta \text{ or } r\sin\theta\cos\theta$

$\left(\frac{dy}{d\theta}=\right)q\sin\theta$

M1

Both correct.

A1

$\frac{dy}{dx}=\frac{-2\sin\theta}{-6\sin 2\theta}$

Use chain rule $\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$ *; condone one slip*

M1

$=\frac{2\sin\theta}{6\times 2\sin\theta\cos\theta}=\frac{1}{6\cos\theta}$

k *= 6 must come from correct working seen* ***AG***

A1

4

(ii) $\theta=\frac{\pi}{3}\quad m_T=\frac{1}{3}$

ft on k $\left(\frac{1}{k\times\frac{1}{2}}\right)$

k *need not be numerical*

B1ft

$m_N=-3$

ft on m_T

B1ft

$(x, y) = \left(-\frac{3}{2}, 1\right)$

B1

Normal $y - 1 = -3\left(x + \frac{3}{2}\right)$

CAO; any correct form, ISW.

$2y + 6x + 7 = 0$

B1

4

(b) $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$p + q\cos 2x$; Allow different letters for x or mixture eg θ even for A1and the following A1ft

M1

A1

$\int p \, dx = px \quad \int q \cos 2x = \frac{1}{2} q \sin 2x$

Both integrals correct;

ft on p and q

A1ft

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \left[\frac{x}{2} - \frac{1}{4}\sin 2x\right]$$

$$= \left(\frac{\pi}{8} - \frac{1}{4}\right) - \left(-\frac{\pi}{8} - \left(-\frac{1}{4}\right)\right)$$

Correct use of limits;

$F\left(\frac{\pi}{4}\right) - F\left(-\frac{\pi}{4}\right)$ *or* $2F\left(\frac{\pi}{4}\right)$

$F(x) = px + r\sin 2x$ *and* $\sin\frac{\pi}{2}, \sin\left(-\frac{\pi}{2}\right)$

must be evaluated correctly for m1

m1

$$= \frac{\pi}{4} - \frac{1}{2}$$

CSO OE ISW

A1

5

Alternative

$$\int \sin^2 x\mathrm{d}x = -\sin x\cos x - \int -\cos x\cos x\mathrm{d}x$$

Use parts; condone sign slips

(M1)

$$= -\sin x\cos x + \int 1 - \sin^2 x\mathrm{d}x$$

Use $\cos^2 x = 1 - \sin^2 x$

(m1)

$$2\int \sin^2 x\mathrm{d}x = -\sin x\cos x + x$$

(A1)

$$2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x\mathrm{d}x = \mathrm{G}\left(\frac{\pi}{4}\right) - \mathrm{G}\left(-\frac{\pi}{4}\right)$$

Correct use of limits

(m1)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x\mathrm{d}x = \frac{\pi}{4} - \frac{1}{2}$$

(A1)

(5)

[13]

Q12.

(a) $(C =)\frac{2}{e}$ or $2e^{-1}$ or $2\left(\frac{1}{e}\right)$ or $2\left(e^{-1}\right)$

One of these answers only.
Not 0.736 but allow ISW.

B1

1

(b) $\frac{\mathrm{d}}{\mathrm{d}x}(2y) = 2\frac{\mathrm{d}y}{\mathrm{d}x}$

B1

$\frac{\mathrm{d}}{\mathrm{d}x}\left(e^{2x}y^2\right) = 2e^{2x}y^2 + e^{2x}2y\frac{\mathrm{d}y}{\mathrm{d}x}$

Product; 2 terms added, one with $\frac{dy}{dx}$ *;*

M1

A1 for each term

A1

A1

$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2 + C\right) = 2x$

B1

$\frac{\mathrm{d}y}{\mathrm{d}x} =$

Solve their equation correctly for $\frac{dy}{dx}$

M1

$\frac{x - e^{2x}y^2}{e^{2x}y + 1}$

Condone factor of 2 in both numerator and denominator.   ISW

A1

7

(c) Evaluate $\frac{\mathrm{d}y}{\mathrm{d}x}$ at $\left(1, \frac{1}{\mathrm{e}}\right)$

Substitute $x = 1$ *and* $y = \frac{1}{e}$ *into numerator of* $\frac{dy}{dx}$ *;*
allow one slip

M1

numerator $= 1 - e^2 e^{-2} = 0 \Rightarrow$ stationary point

Conclusion required; must score full marks in part (b)

Allow 1 – 1 = 0 or 2 – 2 = 0

A1

2

[10]

Q13.

$x^2 + xy = e^y$

$2x + y + x\dfrac{dy}{dx} = e^y \dfrac{dy}{dx}$

$2x$

B1

Use product rule

M1A1

RHS

B1

$(-1, 0)$ $\dfrac{dy}{dx} = -1$

CSO

A1

[5]

Q14.

(a) (i) $\sin 2\theta = 2 \sin \theta \cos \theta$

B1

$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

OE condone use of x etc, but variable must be consistent

B1

2

(ii) $\sin \theta = \dfrac{4}{5} \Rightarrow \sin 2\theta = 2 \times \dfrac{4}{5} \times \dfrac{3}{5} = \dfrac{24}{25}$

AG

Use of 106.26° …. B0

B1

or

$2 \times \sin\left(\cos^{-1}\frac{3}{5}\right) \times \frac{3}{5}$

$\cos 2\theta = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$

–0.28

B1

2

(b) (i) $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 6\cos 2\theta$, $\frac{\mathrm{d}y}{\mathrm{d}\theta} = -8\sin 2\theta$

Attempt both derivatives. ie $p \cos 2\theta$

M1

Both correct. $q \sin 2\theta$

A1

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{4}{3}\frac{\sin 2\theta}{\cos 2\theta}$ ISW

CSO *OE*

A1

3

(ii) $P\left(\frac{72}{25}, -\frac{28}{25}\right)$

(2.88, –1.12)

B1F

Gradient $= -\frac{4}{3} \times -\frac{24}{7}$

Their $\frac{q\sin 2\theta}{p\cos 2\theta}$ **or** $\frac{p\cos 2\theta}{q\sin 2\theta}$

must be working with rational numbers

M1

Tangent $y + \frac{28}{25} = \frac{32}{7}\left(x - \frac{72}{25}\right)$ ISW

Any correct form.

7y = 32x – 100

Fractions in simplest form

Equation required

A1

3

[10]

Q15.

(a) $\frac{dx}{dt} = -3 \quad \frac{dy}{dt} = 6t^2$

Both derivatives correct; PI

B1

$\frac{dy}{dx} = -\frac{6t^2}{3}$

Correct use of chain rule

M1

$= -2t^2$

CSO

A1

3

(b) $t = 1 \quad m_T = -2 \quad m_N = \frac{1}{2}$

Substitute $t = 1$ $m_N = -\frac{1}{m_T}$

M1

F on gradient; $m_T \neq \pm 1$

A1F

Attempt at equation of normal using $(x, y) = (-2, 3)$

Condone one error

M1

Normal has equation $y - 3 = \frac{1}{2}(x + 2)$

CSO; ACF

A1

4

(c) $t = \frac{1-x}{3}$ or $t = \sqrt[3]{\frac{y-1}{2}}$

Correct expression for t in terms of x or y

M1

$y = 1 + 2\left(\frac{1-x}{3}\right)^3$

ACF

A1

2

[9]

Q16.

(a) $x^3 + \cos\pi = 7 \Rightarrow x^3 - 1 = 7$

Or $x = \sqrt[3]{7 - \cos\pi}$

M1

$x = 2$

CSO

A1

2

(b) $\frac{d}{dx}(x^3 y) = 3x^2 y + x^3 \frac{dy}{dx}$

2 terms added, one with $\frac{dy}{dx}$

M1A1

$\frac{d}{dx}(\cos \pi y) = -\pi \sin(\pi y)\frac{dy}{dx}$

B1

At $(2,1)$ $\quad 3 \times 4 + 8\frac{dy}{dx} - \pi \sin\pi \frac{dy}{dx} = 0$

Substitute candidate's x from (a) and
y = 1 with 0 on RHS and both derivatives attempted and no extra derivatives

M1

$\frac{dy}{dx} = -\frac{3}{2}$

CSO; OE

A1

5

[7]

Q17.

(a) $x^2 \frac{dy}{dx} + 2xy$

Product rule used. Allow 1 error

M1A1

$+ 3y^2 \frac{dy}{dx}$

Chain rule

B1

= 2

RHS and equation with no spurious $\frac{dy}{dx}$ *unless recovered.*

B1

$(2, 1), \quad 4\frac{dy}{dx} + 4 + 3\frac{dy}{dx} = 2$

Substitute (2, 1)

M1

$\frac{dy}{dx} = -\frac{2}{7}$

CSO

A1

6

(b) $\frac{dy}{dx} = 0 \Rightarrow$

Derivative = 0 used

M1

$xy = 1$

OE

A1

$x^2 \times \frac{1}{x} + \frac{1}{x^3} = 2x + 1$

Use $xy = k$ to eliminate y on LHS

m1

$\frac{1}{x^3} = x + 1$

Answer given; CSO

A1

4

[10]

Q18.

(a) $\frac{dx}{dt} = \frac{1}{t^2} \quad \frac{dy}{dt} = 1 - \frac{1}{2t^2}$

B1B1

$\frac{dy}{dx} = \frac{1 - \frac{1}{2t^2}}{-\frac{1}{t^2}} \quad \left(= \frac{2t^2 - 1}{-2} \right)$

Their $\frac{\frac{dy}{dt}}{\frac{dx}{dt}}$; condone 1 slip

M1

CSO; ISW

A1

Alternative

$y = \frac{1}{x} + \frac{x}{2}$

(B1)

$\frac{dy}{dx} = -\frac{1}{x^2} + \frac{1}{2}$

(B1)

Substitute $x = \frac{1}{t}$

(M1)

$\frac{dy}{dx} = -t^2 + \frac{1}{2}$

CSO

(A1)

4

(b) $t = 1 \quad \frac{dy}{dx} = -\frac{1}{2}$

Substitute $t = 1$ *in* $\frac{f(t)}{g(t)} \neq k$

M1

$m_T = -\frac{1}{2} \Rightarrow m_N = 2$

F on $m_T \neq 0$; if in t → numerical later

B1F

$(x, y) = (1, \frac{3}{2})$

$PI\ \frac{3}{2} = m(\times 1) + c$

B1

$(y - \frac{3}{2}) = 2(x - 1)$ **or** $y = 2x + c, c = -\frac{1}{2}$

ISW, CSO (a) and (b) all correct

A1

4

(c) $y = \frac{1}{\frac{1}{t}} + \frac{1}{2} \times \frac{1}{t}$

Attempt to use $t = \frac{1}{x}$ *to eliminate t*

t, or equivalent

M1

$= \frac{1}{x} + \frac{x}{2}$

A1

$2xy = 2 + x^2 \Rightarrow x^2 - 2xy + 2 = 0$

Correct algebra to AG with k = 2
allow k = 2 stated k = 2,

no working or from $\left(1, \frac{1}{3}\right)$: *0/3*

A1

Alternative **or**

$\left(\frac{1}{t}\right)^2 - 2\left(\frac{1}{t}\right)\left(t + \frac{1}{2t}\right) = -2$ | $xy = \frac{1}{t}\left(t + \frac{1}{2t}\right) = 1 + \frac{x^2}{2}$

Substitute and multiply out

(M1)

Eliminate t

(A1)

$\Rightarrow x^2 - 2xy + 2 = 0$

Conclusion, k = 2

(A1)

3

[11]

Q19.

$8x + 2y\frac{dy}{dx} = 3y + 3x\frac{dy}{dx}$

$8x$ and 4 → 0

B1

$2y\frac{dy}{dx}$

B1

$3y + 3x\frac{dy}{dx}$

Two terms with one $\frac{dy}{dx}$

M1A1

at (1, 3) (gradient) $\frac{dy}{dx} = \frac{1}{3}$

CSO

A1

[5]

Q20.

(a) (i) $t = \frac{1}{2} \quad x = 2 \times \frac{1}{2} + \frac{1}{\left(\frac{1}{2}\right)^2} \quad y = 2 \times \frac{1}{2} - \frac{1}{\left(\frac{1}{2}\right)^2}$

M1

$x = 5 \qquad y = -3$

A1

2

(ii) $\frac{dy}{dt} = 2 + 2t^{-3} \quad \frac{dx}{dt} = 2 - 2t^{-3}$

2 and $\frac{d}{dt}\left(\frac{1}{t^2}\right)$ *attempted in both derivatives*

M1A1

$t = \frac{1}{2} \quad \frac{dy}{dx} = \frac{2 + \frac{2}{1/8}}{2 - \frac{2}{1/8}} = -\frac{9}{7}$

use chain rule; expressions can be in terms of t or evaluated

M1

CAO or any equivalent fraction (not decimals)

A1

$y + 3 = -\frac{9}{7}(x - 5)$

ft on, x y and gradient

if $y = mx + c$ used, c must be found correctly and the equation must be rewritten

B1F
5

(b) $x - y = \frac{2}{t^2}$ $x + y = 4t$

either correct expression

or both of $x - y = 4t$ and $x + y = \frac{2}{t^2}$

M1

$$\frac{2}{(x-y)} = \left(\frac{x+y}{4}\right)^2$$

eliminate t

M1

$32 = (x - y)(x + y)^2$

or $(x - y)(x + y)^2 = \frac{2}{t^2} \times (4t)^2 = 32$

$k = 32$ alone, no marks

A1
3

[10]

Q21.

$$3x\frac{dy}{dx} + 3y - 4y\frac{dy}{dx} = 0$$

attempt implicit differentiation

M1

product

A1

chain

A1

constant

B1

$$\frac{dy}{dx} = -\frac{3}{2}$$

CSO

A1

ALTERNATIVE METHOD

$x = \frac{2}{3}y + \frac{4}{3y}$

solve for x = expression in y and differentiate with respect to y

(M1)

$\frac{dx}{dy} = \frac{2}{3} - \frac{4}{3y^2}$

(A1A1)

$y = 1, \frac{dx}{dy} = \frac{2}{3} - \frac{4}{3}$

substitute y = 1

(M1)

$\frac{dx}{dy} = -\frac{3}{2}$

CSO

(A1)

[5]

Q22.

(a) $\frac{dx}{dt} = 4 \quad \frac{dy}{dt} = -\frac{1}{2t^2}$

differentiate. 4; at^{-2} seen

M1

both derivatives correct

A1

$\frac{dy}{dx} = -\frac{1}{2t^2} \times \frac{1}{4}$

use chain rule candidates' $\frac{dy}{dt} / \frac{dx}{dt}$

M1

$t = \frac{1}{2} \quad \frac{dy}{dx} = -\frac{1}{2}$

CSO

A1

ALTERNATIVE METHOD:

$y = \frac{2}{x-3} - 1$ and differentiate

differentiate expression in y and x

M1

$\frac{dy}{dx} = \frac{-2}{(x-3)^2}$

correct

A1

$x = 5$

$\frac{dy}{dx} = \frac{-2}{(5-3)^2}$

find and therefore use x (and y)

m1

$\frac{dy}{dx} = -\frac{1}{2}$

A1

4

(b) gradient of normal = 2

F if gradient ≠ ±1

B1F

$(x, y) = (5, 0)$ $\frac{y}{x-5} = 2$

calculate and use (x, y) on normal

M1

F on gradient of normal ACF

A1F

3

(c) $x - 3 = 4t$ or $y + 1 = \frac{1}{2t}$

or $t = \frac{x-3}{4}$ *or* $\frac{1}{t} = 2(y+1)$

B1

$(x - 3)(y + 1) = 2$

eliminate t; allow one error

M1

$y = \frac{1}{2(x-3)} - 1$ *ACF*

accept $\frac{2(x-3)}{4}$

SC allow marks for part (c) if done in part (a)

A1

3

[10]

Q23.

(a) (i) $\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = -8t$

CAO

B1, B1

2

(ii) $\frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} = \frac{-8t}{2} = -4t$

M1

Chain rule in correct form

*ft on sign coefficient errors (**not** power of t)*

A1F

2

(b) $m_T = -4, \quad m_N = \frac{1}{4}$

ft on $\frac{dy}{dx}$ *if f(t)*

B1F,
B1F

$x = 3 \quad y = -3$

$\frac{y - -3}{x - 3} = \frac{1}{4} \Rightarrow \frac{y + 3}{x - 3} = \frac{1}{4}$

Use candidate's (x, y) and m_N

M1

Any correct form; ISW; CAO

A1

4

(c) $t=\frac{x-1}{2}$

M1

$y=1-4\left(\frac{x-1}{2}\right)^2$

Substitute for t
Simplification not required but CAO
Or equivalent methods / forms:

$y=2x-x^2$, $t^2=\frac{1-y}{4}$,

$\left(\frac{x-1}{2}\right)^2=\frac{1-y}{4}$

M1A1

3

[11]

Q24.

(a) $x = 1$, $5a^2 - a - 4 = 0$

condone y for a

M1

$(5a+4)(a-1)=0$, $a=1$

AG – be convinced, both factors seen

or $a=-\frac{4}{5}$ *or 1* $\Rightarrow$ $a = 1$
A0 for 2 positive roots
(substitute (1, 1) $\Rightarrow$ *5 = 5 no marks)*

A1

2

(b) $\frac{dy}{dx}+4$

(Ignore ' $\frac{dy}{dx}$ *=' if not used, otherwise loses final A1)*

B1B1

$= 10xy^2 + 10x^2 y\frac{dy}{dx}$

attempt product rule, see two terms added

M1

chain rule, $\frac{dy}{dx}$ *attached to one term only*

M1

condone 5 × 2 for 10

A1

$x = 1, y = 1 \qquad \frac{dy}{dx} + 4 = 10 + 10\frac{dy}{dx}$

two terms, or more, in $\frac{dy}{dx}$

M1

$\frac{dy}{dx} = -\frac{6}{9} = \left(-\frac{2}{3}\right)$

CSO

A1 7

Alt (for last two marks)

$\frac{dy}{dx} = \frac{10xy^2 - 4}{1 - 10x^2 y}$

find $\frac{dy}{dx}$ *in terms of* x, y *and substitute* x *= 1,*
y *= 1 must be from expression with two terms or more in* $\frac{dy}{dx}$

(M1)

(1, 1) $\Rightarrow \frac{10-4}{1-10} = -\frac{6}{9}$

(A1)

(c) $\frac{y-1}{x-1} = -\frac{2}{3}$ (OE)

ft on gradient

ISW after any correct form

B1F

1

[10]

Q25.

(a) (i) $\frac{dx}{d\theta} = -\sin\theta \quad \frac{dy}{d\theta} = 2\cos 2\theta$

B1 B1

2

(ii) $\frac{dy}{dx} = -\frac{2\cos 2\theta}{\sin\theta}, \quad \frac{dy}{dx} = -\frac{2\cos\frac{\pi}{3}}{\sin\frac{\pi}{6}} = -2$

use chain rule $\text{their}\frac{dy}{d\theta} \Big/ \text{their}\frac{dx}{d\theta}$ *and*

M1

substitute $\theta = \frac{\pi}{6}$

A1

2

(b) $y = 2\sin\theta\cos\theta = 2\sqrt{1-\cos^2\theta}\,\cos\theta$

use sin 2θ = 2sin θ cos θ

B1

use sin² θ = 1 − cos² θ

B1

$y = 2\sqrt{1-x^2}\,x$

sin θ, cos θ in terms of x

M1

$y^2 = 4x^2(1-x^2)$

all correct CSO

A1

4

Alt

$y^2 = \sin^2 2\theta = (2\sin\theta\cos\theta)^2$

use of double angle formula

(B1)

$= (4)\sin^2\theta\cos^2\theta = (4)(1 - \cos^2\theta)\cos^2\theta$

use of $s^2 + c^2 = 1$ to eliminate $\sin\theta$

(B1)

$= (4)(1 - x^2)x^2$

Substitute $\cos\theta$ for x

(M1)

$= 4(1 - x^2)x^2$

CSO

(A1)

(4)

[8]

Q26.

(a) $\frac{dy}{dt} = \frac{-2}{t^2}$ $\quad \frac{dx}{dt} = -4$

M1A1

$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = \frac{1}{2t^2}$

Use chain rule

m1

Follow on use of chain rule (if f (t))

A1F

***Or** eliminate t : M1 $y = f(x)$ attempt to differentiate M1A1 chain rule*
A1F reintroduce t

4

(b) $t = 2 \quad m_T = \frac{1}{8}$

follow on gradient (possibly used later)

B1F

$x = -5 \quad y = 2$

B1

$y - 2 = \frac{1}{8}(x + 5)$

Their (x, y), m

M1

$x - 8y + 21 = 0$

Ft on (x, y) and m

A1F

4

(c) $x - 3 = -4t \qquad y - 1 = \frac{2}{t}$

PI

M1

$(x - 3)(y - 1) = -4t \times \frac{2}{t} = (-8)$

Attempt to eliminate t

M1

AG convincingly obtained

A1

3

[11]

Q27.

(a) $x = 1 \qquad y^2 - y + 3 - 5 = 0$

M1

$(y - 2)(y + 1) = 0$

Attempt to solve quadratic equation with x = 1

M1

$y = 2 \quad y = -1$

A1

3

(b) (i) $2y\frac{dy}{dx} - x\frac{dy}{dx} - y + 6x = 0$

$+6x$; $-5 \to 0$

B1B1

Chain rule

B1

Product rule (M1 two terms)

M1A1

$6x - y + (2y - x)\dfrac{dy}{dx} = 0$

Factorise and obtain answer given

A1

6

Alternative

$\dfrac{dy}{dx}(y - x)^2 = (y - x)(0 - 6x)$

$5 \to 0$

(B1)

$-6x$

(B1)

$-(5 - 3x^2)\left(\dfrac{dy}{dx} - 1\right)$

Recognisable attempt at quotient rule

(M1)

Completely correct OE

(A1)

$\dfrac{dy}{dx}\left[(y + x)^2 + (5 - 3x^2)\right] = (y - x)(-6x)$

$+ (5 - 3x^2)$

Factorise out $\dfrac{dy}{dx}$

(A1)

Given answer

Correct answer from correct working

Be convinced

(A1)

(ii) $(1, 2)$ $\quad \frac{dy}{dx} = -\frac{4}{3}$

Substitute x = 1 and one y value from (a)

M1

$(1, -1)$ $\quad \frac{dy}{dx} = \frac{7}{3}$

Both; follow on candidates y s

OE $\frac{-7}{-3}$ *; 3SF*

A1F

2

(iii) $y - 6x = 0$

B1

$(6x)^2 - x \times 6x + 3x^2 - 5 = 0$

M1

$36x^2 - 6x^2 + 3x^2 - 5 = 0$

$33x^2 - 5 = 0$

AG convincingly obtained

A1

3

[14]

Q28.

(a) $t = \frac{1}{2}$ $\quad x = 3 \quad y = 2$

B1B1

2

(b) $t = \frac{1}{y}$ $\qquad x = 2\frac{1}{y} + y$

Attempt to eliminate t

***SC** verification using t 1/2*

M1

$xy = 2 + y^2 \qquad xy - y^2 = 2$

AG; convincingly found

A1

2

(c) Implicit differentiation: $x\frac{dy}{dx} + y, -2y\frac{dy}{dx} = 0$

Attempt at equation with $\frac{dy}{dx}$ *but not "* $\frac{dy}{du} = \ldots$ *"*

M1
A1A1

RHS = 0

B1

At (3, 2) $3\frac{dy}{dx} + 2 - 4\frac{dy}{dx} = 0$

Use of (3, 2)

m1

$\frac{dy}{dx} = 2$

AG; convincingly obtained

A1

6

$$\frac{dy}{dx} = \frac{dy}{dt}\frac{1}{\frac{dx}{dt}} = -\frac{1}{t^2}\frac{1}{2-\frac{1}{t^2}}$$

Parametric differentiation:

Attempt chain rule, PI

(M1)
(A1A1)

$\frac{1}{t^2} = y^2$

Or sub t = $\frac{1}{2}$ *; –4 in numerator is sufficient for this*

(B1)

$y = 2$ $\frac{dy}{dx} = \frac{-4}{2-4} = 2$

AG; convincingly obtained

(M1A1)

(6)

OR

$x = y + 2y^{-1}$ M1

$\frac{dx}{dy} = 1 - 2y^{-2}$ M1A1A1

$= 1 - \frac{1}{2} = \frac{1}{2}$ B1 $\to \frac{dy}{dx} = 2$ A1

[10]